



Lesson 7.2 — Radians and Degrees

Big idea: A radian is the angle whose arc length equals the radius. Radians are the natural unit for math; degrees are convenient for everyday measurement. Convert with the constant $\pi = 180^\circ$.

Quick review

Conversion: π radians = 180° . So $1 \text{ rad} = 180/\pi^\circ \approx 57.30^\circ$. $1^\circ = \pi/180 \text{ rad}$.

Key conversions:

$0^\circ = 0$, $30^\circ = \pi/6$, $45^\circ = \pi/4$, $60^\circ = \pi/3$, $90^\circ = \pi/2$, $180^\circ = \pi$, $270^\circ = 3\pi/2$, $360^\circ = 2\pi$.

Why use radians: arc length $s = r\theta$ and area $A = (1/2)r^2\theta$ both require θ in radians.

To convert: multiply degrees by $\pi/180$; multiply radians by $180/\pi$.

Worked example

Worked example. Convert 135° to radians and $\pi/3$ to degrees.

$$135^\circ \cdot \pi/180 = 135\pi/180 = 3\pi/4. \quad \pi/3 \cdot 180/\pi = 60^\circ.$$

Warm-ups

1. Convert 90° to radians.
2. Convert $\pi/6$ to degrees.
3. Convert 360° to radians.

Practice

1. Convert 210° to radians.
2. Convert $5\pi/6$ to degrees.
3. An arc of radius 4 cm subtends an angle of $\pi/3$. Find the arc length.

Challenge

1. A circle has radius 10 in. The minute hand sweeps 25 min. Find the radian measure of the angle, then the arc length.
2. Convert 1 rad to degrees, minutes, and seconds.



Answer key — Lesson 7.2

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $\pi/2$.
2. 30° .
3. 2π .

Practice

1. $210 \cdot \pi/180 = 7\pi/6$.
2. 150° .
3. $s = r\theta = 4 \cdot \pi/3 = 4\pi/3 \text{ cm} \approx 4.19 \text{ cm}$.

Challenge

1. 25 min out of 60 = $25/60$ of full revolution = $(25/60)(2\pi) = 5\pi/6$ rad. Arc length = $10 \cdot 5\pi/6 = 25\pi/3 \approx 26.18$ in.
2. $1 \text{ rad} \approx 57.2958^\circ$. $0.2958^\circ \cdot 60 = 17.7480' \rightarrow 17' + 0.7480' \cdot 60 \approx 44.88''$. So $\approx 57^\circ 17' 45''$.